Laboratory 03

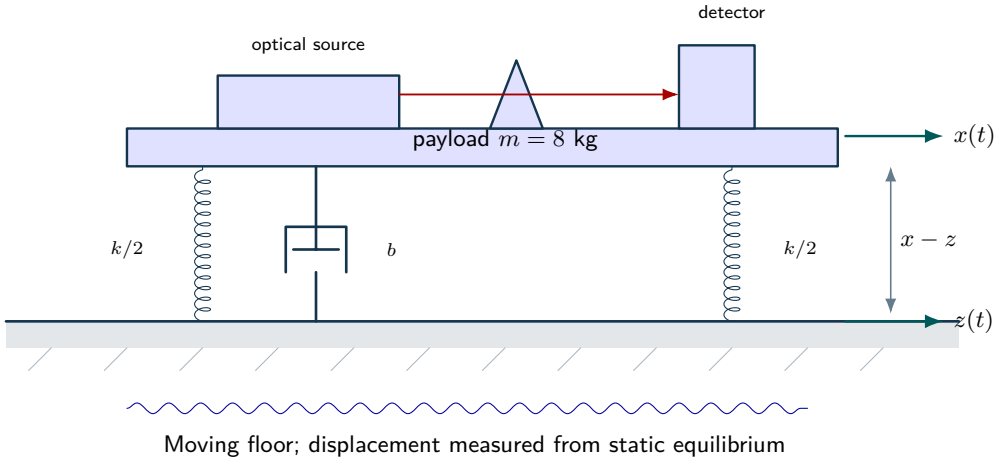
A precision platform under floor motion

Worked solutions – five integrated problems

Evidence boundary. Original hypothetical scenarios. Derive, calculate and justify decisions; every numerical record is synthetic. Figures are conceptual and not to scale. **Shared nominal data.** Nominal payload: $m = 8$ kg, $k = 3200$ N/m and $\zeta = 0.08$ unless varied. Absolute payload motion is x; floor motion is z. Spring and damper depend on relative motion about static equilibrium.

1. An optical platform on a moving foundation

A precision payload is supported by a spring and damper attached to a moving floor. Positions are measured about the loaded static equilibrium, so gravity does not appear in the incremental motion equation.



An optical payload rests on parallel spring and damping elements. The two springs each have stiffness $k/2$, giving total k. Both the floor and payload move; spring extension and damping velocity are relative quantities. Weight is balanced at the static equilibrium about which the model is written.

(a) An 8 kg platform is attached to a moving floor by a spring $k = 3200$ N/m and damper b. Let x and z be absolute platform and floor displacements. Derive X/Z and the relative-displacement transfer $(X - Z)/Z$.

Worked reasoning. Forces depend on relative deformation and velocity: $m\ddot{x} = -b(\dot{x} - \dot{z}) - k(x - z)$. Therefore $X/Z = (bs + k)/(ms^2 + bs + k)$. Subtracting one gives $(X - Z)/Z = -ms^2/(ms^2 + bs + k)$. Replacing the moving floor by an applied force would change the numerator and the isolation interpretation; the two input models are not interchangeable.

(b) Find ω_n and the damping coefficients for ratios 0.08 and 0.4. Calculate the closed-form poles for both cases.

Worked reasoning. The natural frequency is $\sqrt{k/m} = 20$ rad/s. From $b = 2\zeta\sqrt{mk} = 320\zeta$, the damping coefficients are 25.6 and 128 N s/m. The poles are $-\zeta\omega_n \pm j\omega_n\sqrt{1 - \zeta^2}$: $-1.6 \pm j\,20\sqrt{0.9936}$ and $-8 \pm j\,20\sqrt{0.84}$. Both are stable underdamped pairs. Increasing damping moves the real parts left while reducing damped frequency.

(c) Find the zero-frequency gains from floor displacement to absolute payload motion and to relative suspension travel. Explain the physical difference and why using the force-input transfer for this experiment would be wrong.

Worked reasoning. With $G_{xz} = (bs + k)/(ms^2 + bs + k)$, $G_{xz}(0) = 1$. Relative travel has $G_{(x-z)z} = G_{xz} - 1 = -ms^2/(ms^2 + bs + k)$, giving zero DC gain. A sufficiently slow floor displacement moves the entire payload without incremental suspension deflection. A separately applied force instead uses $G_{xf} = 1/(ms^2 + bs + k)$, with different units and numerator; floor displacement is not a force.

2. Damping a resonance while preserving isolation

The design must cross a resonant region during operation but should isolate higher-frequency floor motion. Compare the competing effects of damping rather than ranking one damper as universally better.

(a) Using $r = \omega/\omega_n$, derive the condition for absolute displacement transmissibility below one. Explain why the nonzero threshold is independent of positive damping in this ideal model.

Worked reasoning. The squared transmissibility is $T^2 = [1 + (2\zeta r)^2]/[(1 - r^2)^2 + (2\zeta r)^2]$. For $T^2 < 1$, subtract the common damping terms: $1 < (1 - r^2)^2$, or $r^2(r^2 - 2) > 0$. Hence isolation begins at $r > \sqrt{2}$, namely $\omega > 20\sqrt{2} \approx 28.284$ rad/s. At $r = 0$ and $r = \sqrt{2}$, $T = 1$. The cancellation follows from this specific base-excitation numerator.

(b) Calculate transmissibility at $r = 1$ for damping ratios 0.08 and 0.4. Which design better suppresses floor motion at that frequency? Do these values necessarily equal the exact peak magnitudes?

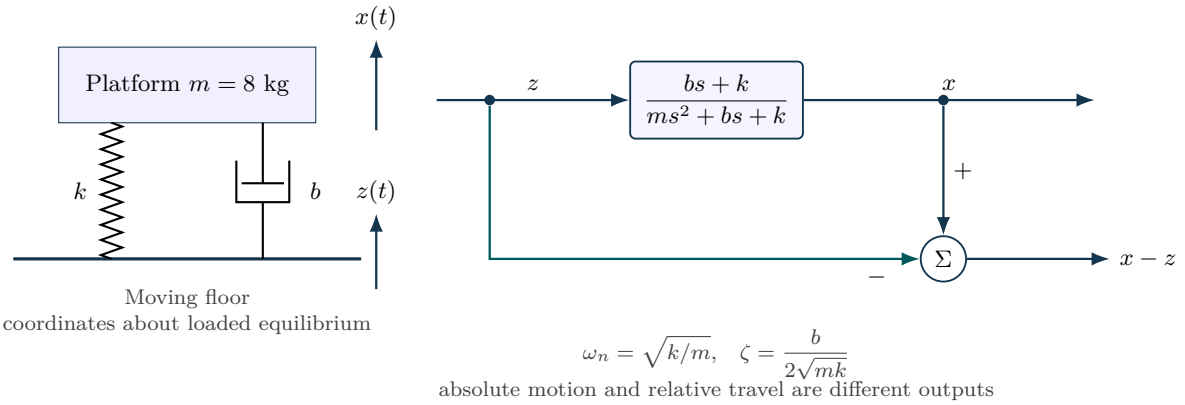
Worked reasoning. At $r = 1$, $T = \sqrt{1 + 4\zeta^2}/(2\zeta)$. The two values are $\sqrt{1.0256}/0.16 \approx 6.3295$ and $\sqrt{1.64}/0.8 \approx 1.6008$. The larger damping transmits less motion at this frequency. The base-excited transfer has a zero, so its exact magnitude maximum need not occur at ω_n. Evaluating at $r = 1$ is a useful comparison, not a proof of the peak frequency.

(c) Analyze the limit $\zeta \to 0^+$ at $r = 1$ and at a fixed $r > \sqrt{2}$. Use the result to explain why an isolation threshold alone is an incomplete design specification.

Worked reasoning. At $r = 1$, the transmissibility is $\sqrt{1 + 4\zeta^2}/(2\zeta)$ and diverges as damping tends to zero. At fixed $r > \sqrt{2}$, its limit is $1/(r^2 - 1) < 1$. Thus a weakly damped system can isolate well above resonance while suffering an arbitrarily large resonant amplification. The threshold $r = \sqrt{2}$ does not constrain the peak, decay time, usable travel or force limits.

3. Absolute vibration and suspension-travel limits

The same floor motion can create different specifications for the optical payload and its supporting suspension. Evaluate both outputs before selecting a mount.



Base excitation drives absolute motion; subtracting the floor coordinate gives relative travel. The spring and damper act in parallel.

(a) Evaluate transmissibility at $r = 3$ for both damping ratios. For nonzero damping, derive the high-frequency asymptote and explain the design conflict with Problem 2(b).

Worked reasoning. Substitution gives $T_{0.08} = \sqrt{1.2304/64.2304} \approx 0.13841$ and $T_{0.4} = \sqrt{6.76/69.76} \approx 0.31130$. At large r and fixed $\zeta > 0$, the leading terms give $T \sim 2\zeta/r$. Lower damping therefore improves sufficiently high-frequency isolation, even though it increases near-resonance amplification. For exactly zero damping the asymptote is instead $1/r^2$, so the zero-damping limit needs separate treatment.

(b) At $r = 2$, floor displacement amplitude is 0.5 mm. With $\zeta = 0.08$, find the amplitude of relative travel $x - z$ and decide whether it stays within a 0.8 mm amplitude limit.

Worked reasoning. The relative transfer magnitude is $r^2/\sqrt{(1-r^2)^2 + (2\zeta r)^2}$. At $r = 2$ it is $4/\sqrt{9 + 0.1024} \approx 1.32581$. Multiplying by 0.5 mm gives approximately 0.66291 mm, below 0.8 mm. This is a steady sinusoidal amplitude test; it does not bound startup transients or arbitrary broadband floor motion.

(c) For a fixed floor-displacement amplitude of 0.5 mm, find the limiting absolute and relative amplitudes as frequency tends to infinity with nonzero damping. Relate these limits to the finite-frequency travel check.

Worked reasoning. As $\omega \to \infty$, $X/Z \sim b/(jm\omega) \to 0$, but $(X - Z)/Z \to -1$. The absolute payload amplitude tends to zero while relative travel tends to 0.5 mm, with opposite phase to the floor. At $r = 2$ the nominal relative amplitude is about 0.66291 mm, so the asymptotic limit alone would underestimate the travel needed at that operating frequency.

4. A heavier payload under a tight static-sag requirement

The payload may change, but static support and vibration isolation must still be achieved by the same spring. Treat these as simultaneous constraints.

(a) Keep $k = 3200$ and $b = 25.6$ but increase total mass to 12 kg. Find the new natural frequency, damping ratio and isolation threshold. Explain why keeping the damper fixed does not keep the damping ratio fixed.

Worked reasoning. The new frequency is $\omega_n = \sqrt{3200/12} \approx 16.3299$ rad/s. The ratio is $\zeta = b/[2\sqrt{mk}] = 25.6/[2\sqrt{38400}] \approx 0.06532$. The isolation threshold becomes $\sqrt{2}\omega_n \approx 23.0940$ rad/s. Damping ratio is a normalized quantity depending on both mass and stiffness; a fixed physical damper produces a smaller ratio as mass increases.

(b) For the original 8 kg platform, calculate static spring deflection under gravity. If deflection may not exceed 3 mm, determine the minimum stiffness and the corresponding natural frequency, using $g = 9.81$ m/s^2.

Worked reasoning. The original deflection is $mg/k = 8(9.81)/3200 = 0.024525$ m, or 24.525 mm. The new requirement gives $k \geq mg/0.003 = 26160$ N/m. At equality, $\omega_n = \sqrt{26160/8} = \sqrt{3270} \approx 57.1839$ rad/s. A stiffer support meets static-travel requirements but raises the isolation threshold. The vibration model is understood around the gravity-loaded equilibrium.

(c) Require gravity sag at most 3 mm and isolation of floor motion at 40 rad/s. Derive mass-independent bounds on ω_n and decide whether any passive spring in this model satisfies both.

Worked reasoning. Static sag is $mg/k = g/\omega_n^2$. The sag requirement gives $\omega_n \geq \sqrt{9.81/0.003} \simeq 57.184$ rad/s. Isolation at 40 rad/s requires $40/\omega_n > \sqrt{2}$, or $\omega_n < 28.284$ rad/s. There is no intersection, regardless of payload mass. A different architecture, preload/load-support arrangement or active solution is required; changing the mass without changing the specification cannot remove this conflict.

5. What a decay test says about energy and excitation

A synthetic ring-down is used to infer damping. Distinguish that unforced test from operation on a moving floor, which can continually inject energy.

(a) In a noise-free synthetic free decay, consecutive positive peaks are 1.5 mm and 0.9 mm, separated by 0.32 s. Infer damping ratio, natural frequency and b if mass is 8 kg. State the assumption behind this inference.

Worked reasoning. Let $\delta = \ln(1.5/0.9) = \ln(5/3)$. Then $\zeta = \delta/\sqrt{4\pi^2 + \delta^2} \approx 0.08103$, $\omega_d = 2\pi/0.32$, and $\omega_n = \omega_d/\sqrt{1-\zeta^2} \approx 19.6997$ rad/s. Finally $b = 2m\zeta\omega_n = 16\delta/0.32 \approx 25.5413$ N s/m. The formula assumes one underdamped linear mode, zero continuing excitation, and peaks one complete cycle apart.

(b) Set $z = 0$. Show that mechanical energy decreases for $b > 0$ and discuss asymptotic stability. Explain why the same autonomous energy argument cannot prove decreasing platform energy for arbitrary moving-floor excitation.

Worked reasoning. Use $E = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2$. Differentiating and substituting $m\ddot{x} + b\dot{x} + kx = 0$ gives $\dot{E} = -b\dot{x}^2 \leq 0$. The only invariant motion within $\dot{x} = 0$ has $kx = 0$, hence $x = 0$; with $m, k, b > 0$ the origin is asymptotically stable. A moving floor supplies external work through spring and damper forces, so it can inject energy. The unforced proof cannot bound a forced response without accounting for that input.

(c) For moving floor coordinate z, use physical stored energy $E = m\dot{x}^2/2 + k(x-z)^2/2$. Derive its power balance and identify the term that prevents an unconditional monotonic-decay claim.

Worked reasoning. Differentiation and $m\ddot{x} = -b(\dot{x} - \dot{z}) - k(x - z)$ give $\dot{E} = -b(\dot{x} - \dot{z})^2 - [b(\dot{x} - \dot{z}) + k(x - z)]\dot{z}$. The first term is nonpositive viscous dissipation. The second is power transferred through the moving support and can have either sign. Setting $\dot{z} = 0$ recovers passive decay, but a ring-down argument cannot bound the response to arbitrary sustained floor motion.